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March 3, 1972
SUPERSEDING
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March 22, 1957

FEDERAL SPECIFICATION

ANODES, NICKEL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for nickel anodes used in electroplating baths.

1.2 Classification.

1.2.1 Types. Nickel anodes shall be of the following types, as specified (see 6.2):

Type I - Carbonized
Type II - Depolarized
Type III - Electrorefined

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

O-H-765	Hydrochloric Acid (Muriatic), Technical
QQ-N-290	Nickel Plating (Electrodeposited)
PPP-B-585	Boxes, Wood, Wirebound

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Federal Specifications: (Continued)

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Federal Standards:

Fed. Std. No. 123	Marking for Domestic Shipment (Civil Agencies)
Fed. Test Method Std. No. 151	Metals; Test Methods

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kandas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-D-6054	Drum, Metal-Shipping and Storage

Military Standards:

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

B162	Standard Specification for Nickel Plate, Sheet and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

Society of Automotive Engineers (AMS) Standards:

5553	Nickel Sheet and Strip, Low (0.02 Max) Carbon, Annealed

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, New York 10001).

3. REQUIREMENTS

3.1 Materials. Anodes shall be of solid nickel. They shall be either cast, rolled or forged to shape, or cut from electrolytic cathodes, castings or rolled plate. The Type I anodes (carbonized) shall be furnished either as cast or rolled, as specified (see 6.2), whereas the Type II anodes (depolarized) shall be only furnished as rolled.

3.2 Size, shape, and method of suspension. The size, shape and method of suspension (when applicable) of the anodes shall be as specified (see 6.2). Bar anodes, of elliptical or oval cross section, shall be furnished with a minimum length of 12 inches, unless otherwise specified (see 6.2). Sheared bars and shapes cut from sheet, are acceptable provided length and width does not exceed four linear inches with a maximum thickness of 1/2 inch. Round rods and pellets shall be acceptable provided the diameter does not exceed 1 inch with a maximum length of 1 inch.

3.3 Chemical composition. The nickel anodes shall meet the chemical requirements shown in Table I.

3.4 Type determination. The type of anode shall be that specified (see 1.2.1 and 6.2), when determined by the procedure detailed in 4.5.2.

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TABLE I

Chemical Composition - Percentage

Element	Type I - Carbonized	Type II - Depolarized	Type III - Electrorefined
Nickel plus cobalt 1/	99.00, Min.	99.00, Min.	99.85, Min.
Iron	0.15, Max.	0.15, Max.	0.02, Max.
Manganese	0.05, Max.	0.05, Max.	0.005, Max.
Copper	0.08, Max.	0.10, Max.	0.02, Max.
Silicon	0.20 to 0.35	0.02, Max.	0.005, Max.
Sulphur	0.02, Max.	0.01, Max.	0.03, Max.
Carbon	0.10 to 0.40	0.02, Max.	0.03, Max.
Oxygen	-	0.20, Max.	-
Total other elements 2/	-	-	0.01, Max.

1/ Cobalt counting as nickel.

2/ Other elements, such as P, AS, Sb, Bi, Pb, Sn and Zinc, individually determined, shall not exceed 0.005 percent each.

3.5 Identification. Each anode, except chunks, pieces, pellets or sheared rods, shall be tagged or permanently identified with the following information.

(a) "Anode, Nickel"
(b) Specification number
(c) Type classification

3.6 Workmanship. Anodes shall be clean and substantially free from cracks, warps, casting laps, inclusions, porosity, ragged edges, surface films such as rolling skin, and other defects which may adversely affect uniform corrosion in service. Anodes shall not be broken.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified in the contract or order (see 6.2), a lot shall consist of all anodes of one type and shape in a single shipment. When practicable, each lot shall be made up of anodes of the same melt, size and weight.

4.3 Quality conformance sampling.

4.3.1 For visual examination and nondestructive tests. Unless otherwise specified (see 6.2), samples for visual examination and nondestructive tests shall be selected at random from each lot in accordance with MIL-STD-105, Level II, Acceptable Quality Level (AQL) 1.5 percent defective.

4.3.2 For chemical analysis. Sample anodes shall be selected as follows:

(a) For sheared rods, bars, pellets and shapes not exceeding four inches in length and width, at least 4 sample anodes shall be selected at random from each lot or from each melt, if the lot consists of more than one melt. If the melt or melts cannot be identified, at least 8 anodes shall be selected at random from each lot.

(b) For shapes other than sheared rods, bars, pellets and shapes not exceeding four inches in length and width, at least 2 sample anodes shall be selected at random from each lot or from each melt, if the lot consists of more than one melt. If the melt or melts cannot be identified, at least 4 anodes shall be selected at random from each lot.

4.3.2.1 Preparation for chemical analysis. A composite sample of at least 6 ounces of chips or drillings shall be obtained by drilling, sawing, milling or machining the sample anodes of 4.3.2 in accordance with

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Fed. Test Method Std. No. 151, Method 111. No lubricant shall be used in the operation. The sample shall be free from dirt, oil, grit and other foreign matter. Chips or drillings shall be treated with a magnet to remove any steel introduced in the preparation of the sample. Anodes which have been drilled or machined for chemical analysis may be included in the lot for shipment.

4.3.2.2 Preparation for spectrochemical analysis. A composite sample shall be prepared as detailed in 4.3.2.1 or rod electrodes shall be prepared from each of the sample anodes of 4.3.2 in accordance with Fed. Test Method Std. No. 151, Method 112. The rod samples shall be prepared by conventional methods, suitable for the correct applications and usages of the techniques employed. No lubricant shall be used in the preparation operation. The sample shall be free from dirt, oil, grit and other foreign matter. Anodes, except for sheared rods or chunks (4 by 4 inches or less), from which electrodes have been obtained may be included in the lot for shipment.

4.3.3 For anode type determination. When specified in the contract or order (see 6.2), compliance with the requirement of 3.4 for type of anode shall be determined by selecting one anode, taken at random from each lot.

4.4 Quality conformance examination.

4.4.1 Anodes. Sample anodes selected in accordance with 4.3.1 shall be visually inspected and measured for conformance to the requirements of 3.2, 3.5 and 3.6. If the number of defective items in any sampling exceeds the acceptance number for the specified sample size, the lot represented by the sample shall be rejected.

4.4.2 Packing and marking. The packing and marking of anodes furnished under this specification shall be inspected for conformance to the applicable requirements of Section 5.

4.5 Quality conformance testing.

4.5.1 Chemical composition. Sample anodes, selected in accordance with 4.3.2 shall be tested in accordance with Method 111 or Method 112 of Fed. Test Method Std. No. 151 to determine conformance to the chemical requirements of Table I. If the sample fails to conform to the requirements of Table I, the lot shall be rejected. Standard methods of analysis shall be used (see 6.3).

4.5.2 Type determination. The sample anode selected in accordance with 4.3.3 shall be partially immersed as the anode in an electrolytic cell with an equal immersed area of nickel plate, sheet or strip

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conforming to ASTM B162 or AMS 5553 or of steel nickel-plated in accordance with QQ-N-290 as the cathode and an aqueous solution of 5 percent by volume of hydrochloric acid conforming to O-H-765 as the electrolyte. An anode current density of 100 amperes per square foot shall be applied for one hour. At the end of this time period, a Type I anode shall be covered with a velvety black carbon coating whereas a Type II or a Type III anode shall have no carbon film and shall have an etched metallic appearance. If the anode is not of the type specified, the lot from which the sample was selected shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing. When practicable, anodes of the same size and shape shall be packed together. Except for sheared rods, pellets and square shapes, anodes shall be packed in boxes. Unless otherwise specified, the gross weight of each box shall not exceed approximately 200 pounds. Square shapes, pellets and sheared rods shall be packed in steel drums. Unless otherwise specified, the gross weight of each drum shall not exceed approximately 480 pounds (see 6.2).

5.1.1 Level A. Anodes shall be packed in overseas type boxes in accordance with PPP-B-601 or PPP-B-621. Square shapes, pellets and sheared rods shall be packed in steel drums in accordance with MIL-D-6054.

5.1.2 Level B. Anodes shall be packed in domestic type boxes in accordance with PPP-B-585, PPP-B-601 or PPP-B-621. Square shapes, pellets and sheared rods shall be packed in steel drums in accordance with MIL-D-6054.

5.1.3 Level C. Anodes shall be packed or otherwise prepared for shipment to insure carrier acceptance and to insure safe delivery to destination at the lowest applicable rate. Containers shall meet, as a minimum, the requirements of rules and regulations applicable to the mode of transportation selected.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order (see 6.2) shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military agencies. In addition to any special markings required by the contract or order (see 6.2), shipping containers shall be marked in accordance with MIL-STD-129.

5.2.3 Special markings. Each container shall be marked with the following:

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(a) Specification Number
(b) Type and nomenclature of the anode, as applicable
(c) Name of contractor
(d) Contract Number or Order Number
(e) Quantity
(f) Gross weight

6. NOTES

6.1 Intended use. Nickel anodes are intended for use in nickel-plating baths.

6.1.1 Type I (carbonized) anodes. Type I (carbonized) anodes are made from carbon-containing nickel ingots. They may be rolled or forged into bars of configuration chosen to maximize the surface area exposed to the plating solution and the article to be plated or cast into permanent molds having the desired configuration. The Type I anodes are especially intended for use in nickel plating solutions with a pH of 4 or below. Rolled Type I anodes are generally preferred for auxiliary anodes placed near areas of low current density.

6.1.2 Type II (depolarize) anodes. Type II (depolarized) anodes contain nickel oxides introduces into the melt from which the ingots were poured. They are usually rolled into bars or plate. The Type II anodes are used with high pH nickel baths (above 4) as they will remain active in which the other type anodes may become wholly or intermittently passive.

6.1.3 Type III (electrorefined) anodes. Type III (electrorefined) anodes are high purity anodes where the maximum permissible concentration of iron and copper in the other types can detract from bright nickel deposits. The anodes are generally manufactured by electrowoning or by electroforming. The anodes as pieces, chunks or pellets are held in titanium baskets during electroplating operations (see 6.4).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Type and nomenclature, as applicable (see 1.2.1 and 3.1)
(c) Size, shape and method of suspension (when applicable) (see 3.2)
(d) Size of lot, if different from 4.2
(e) Sampling procedure, if different from 4.3.1
(f) Anode type determination required (see 4.3.3)
(g) Gross weight of packing, if different from 5.1
(h) Applicable level of packing (see 5.1.1, 5.1.2 and 5.1.3)
(i) Additional marking, if necessary (see 5.2)

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6.3 Methods of chemical analysis. It is intended that the chemical analysis be conducted so that no particular method of analysis be specified. As a guide, the contents in anodes may be determined in accordance with ASTM E39, Standard Method for Chemical Analysis of Nickel; ASTM E107, Chemical Analysis of Electronic Nickel and ASTM E129, Spectro Chemical Analysis of Thermionic Nickel Alloys by the Powder-D-C Arc Technique. Metallic impurities in nickel anodes may be more easily determined by the spectrographic methods or photometric methods embodied in the above documents than by wet chemical procedures.

6.4 Suspension. In commercial practice, nickel anodes are usually cast, rolled or forged to an elliptical or oval cross section or simply cuts from pieces of electrowon sheet (metal recovered from an ore by means of electrochemical processing). The elliptical or oval cross sectional anodes are usually provided with one end drilled and tapped to receive a suspension hook. Flat or plate type anodes may be obtained, provided with holes near each of three corners or with two lugs extending upward from the top edge with a hole in each for suspension in the bath. Small chunks (1 to 2 inch squares) are not usually provided with suspension hooks as they are held in baskets made of perforated or pierced and expanded titanium sheet or in bags made of cotton or synthetic fibers.

6.4.1 Bags. Bags are always used with Type II anodes. In some large installations, Type I anodes should be used without bags as this type anode forms a gelatinous membrane around the nickel that effectively prevents solid particles from escaping.

Military Custodians
Army - WC
Navy - AS
Air Force - 84

Review Activities
Army - None
Navy - None
Air Force - None
Defense Supply Agency - GS

User Activities
Army - None
Navy - OS, SH
Air Force - None

Civil Agencies Interest
GSA - FSS

Preparing Activity
Navy - AS
(Proj. No. 3426-0010)

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